

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (Communication System) Examination

Dec 2012

EC710 Digital Image and Speech Processing

Date: 12.12.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the various fields that use the Digital Image Processing. [4]
(b) Explain the adjacency, connectivity and region with an example. [3]
Q - 2 (a) Given a histogram, see what happens when we equalize it twice. [7]

Gray level	0	1	2	3
n_k	50	30	10	10

- (b) From the definition of Fourier transform, prove that following two expressions are Fourier pair (scaling property). [7]

$$f(ax, by) = \frac{1}{|a||b|} F(u/a, v/b)$$

OR

- Q - 2 (a) Below is a histogram of 4-bit image. [7]

r	0	1	2	3	4	5	6	7
H(r)	0	0	50	60	50	20	10	0

Perform the histogram starching so that new image has dynamic range of [0 7].

- (b) Suppose that you form a low pass spatial filter that averages the four immediate neighbors of point (x,y) but excludes the point itself. [7]

(a) Find the equivalent filter H(u,v) in the frequency domain.

(b) Show that result is low pass filter.

- Q - 3 (a) Explain the minimum mean square error filtering. [7]
(b) Explain the color image smoothing and sharpening [7]

OR

- Q - 3 (a) Explain the various color models. [7]
(b) What is the advantage of Homomorphic filtering? Justify this advantage. [7]

SECTION – II

- Q - 4 (a) List the properties of Z-transform with their equivalent Z-transform equation. [3]
(b) What are the representations of digital speech signals? Explain the typical speech communication applications. [4]

- Q - 5 (a) Compare mid riser and mid tread class of uniform quantization. Draw both the classes. [7]
(b) What is the limitation of DPCM? Explain the method to overcome that limitation. [7]

OR

- Q - 5 (a) The short-time average zero-crossing rate defined as [7]

$$Z_n = \frac{1}{2N} \sum_{m=n-N+1}^n |\text{sgn}[x(m)] - \text{sgn}[x(m-1)]|$$

Show that Z_n can be expressed as

$$Z_n = Z_{n-1} + \frac{1}{2N} \{ |\text{sgn}[x(n)] - \text{sgn}[x(n-1)]| - |\text{sgn}[x(n-N)] - \text{sgn}[x(n-N-1)]| \}$$

- (b) Describe the any two time domain methods for speech processing with mathematical expression of each. [7]
Q - 6 (a) Describe procedure of Filter bank design using IIR filters. [7]
(b) What is the principle of linear predictive analysis? [7]

OR

- Q - 6 (a) Explain the main difference between waveform-based coders, vocoders, and hybrid coders for speech coding, in terms of techniques used and the bit-rate/quality range of each. [7]
(b) Describe: "Pitch period estimation using an autocorrelation function. [7]
